AHMED LOAY ELATTAR

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Professional Summary

Mechatronics and robotics engineer with a strong record of building real systems under real constraints — from a factory AGV deployed at Nissan to an award-winning ROV communication architecture. Proficient in embedded systems, PCB design, STM32, ESP32, ROS 2, Nav2/SLAM, and CAN bus.

Professional Experience

- **Procter & Gamble (P&G)**, Egypt *September 2026 – Present*
  - **Innovation Quality Assurance Intern (Baby Care)**
    - Built a Microsoft 365 Copilot agent that automates a monthly eCommerce ratings-and-reviews report, cutting about a week of manual work per report: processes 200,000+ reviews per run through a fixed rulebook and Python analysis, then renders a branded LaTeX PDF with an integrity check.
    - Redesigning it as a question-driven tool for any P&G brand, adding competitor insight, data-quality and authenticity flags, and complaint routing.
    - Automating SOP review in Power Automate: routes each global SOP update to the affected local owner, collects the reply, and tracks revision completion.
    - Installing three end-of-line laser sensors for pack dimensions, feeding the PLC, historian, and Proficy.
- **LEONI**, Egypt *August 2026 – September 2026*
  - **Operational Excellence Intern (Rising Stars Program)**
    - Replaced a manual, hourly paper form for tracking line labour-efficiency (KOSU) with a live Power BI dashboard that reads the plant's existing barcode-scan data and reproduces the form live, with the same colour-coded threshold bands, filterable by date, line, and shift.
    - Built the data model over 140,000+ harness scans: shift and hour derived from timestamps, all DAX measures, and one central threshold table driving colours and trend charts. Built against a read-only traceability database, ready to point at the live SQL source; learned SQL and Power BI from scratch and shipped in 8 days.
    - Ran a 6S audit of the packaging area, tracing recurring findings to one root cause in the rack-labelling standard.
- **Siemens Industrial LLC**, Egypt *July 2026 – August 2026*
  - **SIMATIC Programming 1 in the TIA Portal (Training)**
    - Independently designed a conveyor sorting program on an S7-1500 with ET200SP I/O and a self-built HMI screen: part position tracked as an explicit ladder-logic state, plus a fault layer that latches conflicting commands into an error state until deliberate reset. Also ran a SINAMICS drive and set up PROFINET.
- **Nissan Motor Corporation**, Egypt *July 2023 – September 2023*
  - **Engineering Intern – MSPO — APW — Maintenance**
    - Led the facility's first AGV project, from research to deployment, in 2 months.
    - Evaluated multiple sensing strategies (IR line-following, computer vision) and track materials (electrical/magnetic tape), and selected a drive system after hands-on comparison of DC, brushless, and hoverboard hub motors on torque and cost — delivered a working prototype for ∼27,000 EGP (∼$870 at the time).
    - The deployed AGV took over half of a worker's material handling per shift, saving ∼4,000 EGP (∼$130) a month per shift; worked across MSPO, APW, and Maintenance in a 10+ engineer team.
- **E-JUST Robotics Club**, New Borg El Arab, Alexandria, Egypt
  - **Chief Technical Officer – Electrical & Embedded Systems (MATE ROV)** *Nov 2022 – Present*
    - Designed the ROV's full electronics stack from scratch: custom PCB, power distribution, harness, and a three-tier architecture of a Raspberry Pi 5 (ROS 2), a gateway STM32 (micro-ROS), and CAN bus nodes.
    - Built a wireless OTA firmware update pipeline that pushes new firmware to any STM32 node inside the sealed, underwater vehicle without opening it — firmware transfers over TCP, stages to SPI flash, and deploys over CAN with CRC-32 verification — eliminating disassembly between runs and cutting pre-dive turnaround time in competition.
    - Built a 6-axis motion controller that automatically splits any movement command across seven thrusters in the optimal ratio, with live balancing feedback from an IMU and depth sensor and a self-tuning mode that calibrates response automatically — giving the pilot precise, stable control at any depth.
  - **Mobile Robotics Lead (RoboCup@Home Egypt)** *Apr 2026*
    - Extended the team's robotics stack from underwater 3D control into autonomous 2D ground navigation — built and tuned a full ROS 2 Nav2 deployment with Cartographer SLAM, AMCL localization, and

lidar + wheel odometry + IMU sensor fusion for robust live mapping and loop closure.
  - Owned deep parameter tuning across costmaps, controller server, planner server, and behavior trees, plus an OpenCV-based detect–reason–act perception loop integrated into the navigation stack — now the club's internal reference for mobile-robotics deployments.

Education

- **Egypt–Japan University of Science and Technology (E-JUST)**, New Borg Al-Arab, Alexandria
  – B.Sc. Mechatronics and Robotics Engineering *Oct 2022 – Feb 2027 (Expected)*
- **Gharbiya STEM High School**, Tanta, El Gharbiya
  – Honors Education System – Mathematics *Sep 2019 – Jul 2022*

Core Skills

- **Languages & Platforms:** C, C++, Python, MATLAB; STM32, ESP32, Raspberry Pi, Jetson Nano
- **Embedded & PCB:** CAN bus, SPI, I2C, UART, micro-ROS; Altium Designer, power electronics, PID tuning
- **Robotics:** ROS 2, Nav2, Cartographer SLAM, AMCL, lidar/odometry/IMU fusion, planner tuning, OpenCV
- **Industrial Automation:** S7-1500, TIA Portal, ladder logic, HMI, PROFINET, SINAMICS, ET200SP
- **Fabrication & CAD:** SolidWorks, Fusion 360, FDM printing, CNC (Roland MonoFab), laser cutting
- **Data & Process:** SQL, Power BI (DAX), Power Automate, Git, Lean manufacturing, 6S

Awards and Achievements

- **Best Communication System Award – MATE ROV Egypt, 13th Year (Awarded by NTRA)** *Apr 2025*
  – The only team of ∼30 with a purpose-built communication architecture; organizers created this award category for the first time in the competition's 13-year history.
- **2nd Place – International Drone Challenge, Military Technical College (∼$620 prize)** *Aug 2024*
  – Designed a lightweight frame, tuned PID in Betaflight, and repaired ESC failures mid-competition in a 9-person team; 2nd of 11 teams, 30,000 EGP prize.
- **6th Place (Endurance) & 14th Place (Dynamic) – EVER 5 Electric Vehicle Competition** *Aug 2024*
  – CTO of the electrical team — designed and managed high- and low-voltage systems over a 6-month build; 23 teams (endurance), 22 (dynamic).
- **"No Pain No Gain" Award (twice) – ARC International (Oct 2024) & MATE ROV Egypt debut** *May 2023*
  – Given to first-time teams with exceptional performance, at both the international and regional level.

Projects

- **Two Self-Built 3D Printers (CoreXY with DIY Multi-Material + Cartesian)** *Jul 2023 – Sep 2025*
  – Built a CoreXY 3D printer from scratch running Klipper, with a heavily modified StealthBurner toolhead, automatic filament cutter (Filametrix), multiple filament sensors, and a custom 5-lane multi-material unit for mid-print filament switching — at a fraction of commercial equivalent cost.
  – Developed all macros, bed leveling logic, and a DIY Klicky docking probe from scratch, plus a full power-loss recovery system that resumes prints from the exact layer after any outage with complete remote access for monitoring and control.
  – Both printers (the earlier Cartesian has custom auto-leveling and power-loss recovery) operate as an active small business — 150+ student projects printed on campus and hundreds of orders shipped across Egypt, sustaining a ∼70% gross margin.
- **Projects APOLLO & NEXUS [Self-Built CNC Machines]** *Oct – Nov 2022*
  – NEXUS: first CNC built, a 600×600 mm modular platform used as pen plotter, mill, or laser engraver to learn toolpathing and calibration.
  – APOLLO: 0.5 kW-spindle router machining aluminum, acrylic, wood, and PCBs — the workshop's main tool for custom PCBs and structural parts.
- **Project SPITFIRE [Long-Running Autonomous UAV Platform]** *2015 – Present*
  – The project that started everything: six airframes over 10+ years, from MultiWii to self-compiled ArduPilot on a Pi Zero; each iteration added a skill every later project grew from.

Volunteering

- **E-JUST Robotics Club** – *Training Lead* (50+ trainees) *Oct 2022 – Present*
- **IEEE E-JUST Student Branch** – *Mentor* (6 teams, firefighting boat contest) *Oct 2023 – Present*